Estimation

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Scientific models

Definition 0.1 (Scientific models)

Definition 0.1 (Scientific models)**. Scientific models** are attempts to describe *physical conditions or changes* that occur in the world and universe around us.

Example 0.1 (Scientific models in epidemiology)

Example 0.1 (Scientific models in epidemiology)**.** Epidemiologists typically study *biological conditions and changes*, such as the spread of infectious diseases through populations, or the effects of environmental factors on individuals.

Models as approximations

> …Essentially, all models are wrong, but some are useful. **However, the approximate nature of the model must always be borne in mind.**

[Box and Draper (1987), p. 424; emphasis added]

See also (Dunn and Smyth 2018, sec. 1.8).

Statistical analysis of scientific models

When we perform statistical analyses, we use data to help us choose between models; specifically, to determine which models best explain those data.

Physical processes do not produce data on their own. Data are only produced when scientists implement an *observation process* (that is, a *scientific study*), which is distinct from the

underlying *physical process*. In some cases, the observation process and the physical process interact with each other; this interaction is called the "observer effect".

To learn about the physical processes we are ultimately interested in, we often need to account for the observation process that produced the data we are analyzing. In particular, if some of the planned observations in the study design were not completed, we will likely need to account for the incompleteness of the resulting data set in our analysis. If we are not sure why some observations are incomplete, we may need to model the observation process in addition to the physical process we were originally interested in. For example, if some participants in a study dropped out part-way through, we may need to investigate why those participants dropped out, as opposed to other participants who completed the study.

These kinds of *missing data* issues are outside the scope of these notes; see Van Buuren (2018) for more details.

Estimands, estimates, and estimators

Estimands

Definition 0.2 (Estimand)

Definition 0.2 (Estimand)**.** An **estimand** is an unknown quantity θ whose value we want to know (Pohl et al. 2021; Lawrance et al. 2020).

Example 0.2 (Mean height of students)

Example 0.2 (Mean height of students)**.** If we are trying to determine the mean height of students at our school, then the *population mean* is our estimand.

In statistical contexts, most estimands are parameters of probabilistic models, or functions of model parameters.

i Notation for estimands

Model parameters and other estimands are often symbolized using lower-case Greek letters: $\alpha, \beta, \gamma, \delta$, etc.

Estimates

Definition 0.3 (Estimate/estimated value)

Definition 0.3 (Estimate/estimated value)**.** In statistics, an **estimate** or **estimated value** $\hat{\theta}$ is an informed guess of an estimand θ's value, computed from the observed data $x_1, \dots, x_n$:

$$\hat{\theta} = \hat{\theta}(x_1, \dots, x_n).$$

Example 0.3 (Mean height of students)

Example 0.3 (Mean height of students)**.** Suppose we measure the heights of 50 randomly sampled students from our school, and their sample mean is 175 cm. We might use 175 cm as an *estimate* of the population mean.

Estimators

Definition 0.4 (Estimator)

Definition 0.4 (Estimator)**.** An **estimator** is the function $\hat{\theta}(\cdot)$ that transforms data into an estimate; applied to a random sample $X_1, \dots, X_n$, it is a random variable:

$$\hat{\theta}(X_1, \dots, X_n).$$

i Estimators are random variables

When an estimator is applied to random variables $X_1, \dots, X_n$ rather than to their observed values, the result $\hat{\theta}(X_1, \dots, X_n)$ is also a random variable.

i Notation for estimators

Estimators are often symbolized by placing a $\widehat{\ }$ ("hat") symbol on top of the corresponding estimand; for example, $\hat{\theta}$.
Usually, their dependence on the data is implicit:

$$\hat{\theta} \stackrel{\text{def}}{=} \hat{\theta}(x_1, \dots, x_n)$$

Example 0.4 (Mean height of students)

Example 0.4 (Mean height of students)**.** Suppose we want to estimate the mean height of students at our school, which we will represent as μ, and we measure the heights of $n = 50$ randomly sampled students as random variables $X_1, \dots, X_n$. Then we could use the function

$$\hat{\mu}(X_1, \dots, X_n) \stackrel{\text{def}}{=} \frac{1}{n} \sum_{i=1}^{n} X_i \stackrel{\text{def}}{=} \bar{X}$$

as an *estimator* to produce an *estimate* $\hat{\mu} = \bar{x}$ of μ.
Another estimator would be just the height of the first student sampled:

$$\hat{\mu}^{(2)}(X_1, \dots, X_n) \stackrel{\text{def}}{=} X_1$$

A third possible estimator would be the mean of all sampled students' heights, except for the two most extreme. Using the order statistics $X_{(1)} \leq X_{(2)} \leq \cdots \leq X_{(n)}$ (the observations sorted in increasing order), this estimator drops $X_{(1)}$ and $X_{(n)}$ and averages the remaining $n - 2$ observations:

$$\hat{\mu}^{(3)}(X_1, \dots, X_n) \stackrel{\text{def}}{=} \frac{1}{n-2} \sum_{i=2}^{n-1} X_{(i)}$$

Which of these estimators is best? The answer depends on how we evaluate them (see Section).

Definition 0.5 (Sampling distribution)

Definition 0.5 (Sampling distribution)**.** The **sampling distribution** of an estimator $\hat{\theta}$ is the probability distribution of $\hat{\theta}(X_1, \dots, X_n)$ over repeated random samples $X_1, \dots, X_n$ from the same population: the distribution of the values the estimator would take across all the samples we might have drawn.

Example 0.5 (Sampling distribution of a sample mean)

Example 0.5 (Sampling distribution of a sample mean)**.** Suppose student heights are $X_i \sim_{\text{iid}} \mathrm{N}(\mu, \sigma^2)$ with standard deviation $\sigma = 10$ cm, and we sample $n = 25$ students. The sample mean $\bar{X}$ is a linear combination of independent Gaussian variables, so it is also Gaussian, with mean $\mathrm{E}[\bar{X}] = \mu$ and variance $\mathrm{Var}(\bar{X}) = \sigma^2/n$. Its sampling distribution is therefore

$$\bar{X} \sim \mathrm{N}(\mu, \sigma^2/25),$$

which has standard deviation $\sigma/\sqrt{25} = 10/5 = 2$ cm. Across repeated samples of 25

students, about 95% of the sample means land within $2 \times 2 = 4$ cm of the true mean μ.

Contrasting estimands, estimates, and estimators

It's helpful to keep in mind the mathematical type of each estimation concept:

- estimands are numbers (or vectors of numbers);
- estimates are also numbers (or vectors);
- estimators are functions; an estimator applied to random data, $\hat{\theta}(X_1, \ldots, X_n)$, is a random variable.

Accuracy of estimators

Accuracy

To determine which estimator is best, we need to define *best*. Accuracy is usually most important, and ease of computation is usually secondary.

Estimation error

Definition 0.6 (Estimation error)

Definition 0.6 (Estimation error)**.** The **estimation error** of an estimate $\hat{\theta}$ of a true value θ is the difference between the estimate and the estimand θ:

$$\varepsilon\left(\hat{\theta}\right) \stackrel{\text{def}}{=} \hat{\theta} - \theta$$

Example 0.6 (Estimation error of a mean height)

Example 0.6 (Estimation error of a mean height)**.** Continuing Example 0.3, suppose the population mean height of students at our school were actually $\mu = 172$ cm. Then the estimate $\hat{\mu} = 175$ cm would have estimation error:

$$\begin{aligned} \varepsilon(\hat{\mu}) &= \hat{\mu} - \mu && \text{(definition of estimation error)} \\ &= 175 - 172 && \text{(substitute the estimate and the estimand)} \\ &= 3 \text{ cm} && \text{(subtract)} \end{aligned}$$

In practice we never observe an estimation error, because we do not know the estimand's value; if we did, we would not need to estimate it.

The accuracy of an estimator has no single, agreed formal definition. The usual measures of accuracy, including the bias, mean squared error, and mean absolute error defined in this section, are all functions of the distribution of the estimator's estimation error.

Predictions, fitted values, residuals, and prediction errors

Definition 0.7 (Prediction)

Definition 0.7 (Prediction)**.** A **prediction** (or **predicted value**) $\hat{y}(x)$ of an outcome Y with covariate value $X = x$ is a number computed from x and from data $(x_1, y_1), \ldots, (x_n, y_n)$ by a rule g, and used as a guess of Y:

$$\hat{y}(x) \stackrel{\text{def}}{=} g(x; (x_1, y_1), \ldots, (x_n, y_n)).$$

Example 0.7 (Predicting with an estimated conditional mean)

Example 0.7 (Predicting with an estimated conditional mean)**.** Suppose a model specifies the conditional mean $\mathrm{E}[Y \mid X = x]$ as a function $\mu(x; \tilde{\theta})$ of a parameter vector $\tilde{\theta}$, and $\hat{\tilde{\theta}}$ is an estimate of $\tilde{\theta}$. The usual prediction (Definition 0.7) of Y at $X = x$ is the estimated conditional mean:

$$\hat{y}(x) = \mu(x; \hat{\tilde{\theta}}).$$

Definition 0.8 (Fitted value)

Definition 0.8 (Fitted value)**.** The **fitted value** $\hat{y}_i$ of observation i is the prediction at its own covariate value x_i:

$$\hat{y}_i \stackrel{\text{def}}{=} \hat{y}(x_i).$$

Definition 0.9 (Residual)

Definition 0.9 (Residual)**.** The **residual** for observation i is its observed outcome minus its fitted value:

$$r_i \stackrel{\text{def}}{=} y_i - \hat{y}_i$$

Definition 0.10 (Prediction error)

Definition 0.10 (Prediction error)**.** The **prediction error** of a prediction $\hat{y}$ (such as

$\hat{y}(x)$ or a fitted value $\hat{y}_i$) of an observed outcome y is the prediction minus the outcome:

$$e \stackrel{\text{def}}{=} \hat{y} - y.$$

Exercise 0.1 (Prediction error and residual of a fitted observation)

Exercise 0.1 (Prediction error and residual of a fitted observation)**.** For an observation i that was used to fit the model, write its prediction error e_i in terms of its residual r_i.

Solution 0.1

Solution 0.1.

$$\begin{aligned} e_i &= \hat{y}_i - y_i && \text{(definition of prediction error)} \\ &= -(y_i - \hat{y}_i) && \text{(factor out } -1\text{)} \\ &= -r_i && \text{(definition of residual)} \end{aligned}$$

Theorem 0.1 (Prediction error is the negative of the residual)

Theorem 0.1 (Prediction error is the negative of the residual)**.** *For an observation i that was used to fit the model, the prediction error of its fitted value is the negative of its residual:*

$$e_i = -r_i.$$

Proof

Proof. This is the solution to Exercise 0.1. □

Corollary 0.1 (The residual is the negative of the prediction error)

Corollary 0.1 (The residual is the negative of the prediction error)**.** *For an observation i that was used to fit the model, its residual is the negative of the prediction error of its fitted value:*

$$r_i = -e_i.$$

Proof

Proof. Multiply both sides of Theorem 0.1 by -1. □

Example 0.8 (Residuals and prediction errors for birth weights)

Example 0.8 (Residuals and prediction errors for birth weights)**.** The `birthwt` data record the birth weight (grams) of 189 babies and their mothers' weight at the last menstrual period (pounds), from a study at Baystate Medical Center in Springfield, Massachusetts (Hosmer and Lemeshow 1989). They ship with the R package `MASS` (`MASS::birthwt`), and this site has a copy at `data/birthwt.csv`.
The code below fits a line predicting birth weight from mother's weight, then compares each residual with the negative of its prediction error.

R

```
birthwt <- read.csv("data/birthwt.csv")
fit <- lm(bwt ~ lwt, data = birthwt)
fitted_bwt <- fitted(fit)
residual <- birthwt$bwt - fitted_bwt
prediction_error <- fitted_bwt - birthwt$bwt
head(data.frame(
  bwt = birthwt$bwt,
  fitted = round(fitted_bwt, 1),
  residual = round(residual, 1),
  minus_prediction_error = round(-prediction_error, 1)
))
#>    bwt fitted residual minus_prediction_error
#> 1 2523 3175.7   -652.7                 -652.7
#> 2 2551 3056.1   -505.1                 -505.1
#> 3 2557 2834.7   -277.7                 -277.7
#> 4 2594 2848.0   -254.0                 -254.0
#> 5 2600 2843.5   -243.5                 -243.5
#> 6 2622 2918.8   -296.8                 -296.8
max(abs(residual - (-prediction_error)))
#> [1] 0
```

Python

```python
import numpy as np
import pandas as pd

birthwt = pd.read_csv("data/birthwt.csv")
X = np.column_stack([np.ones(len(birthwt)), birthwt["lwt"]])
y = birthwt["bwt"].to_numpy(dtype=float)
coef = np.linalg.lstsq(X, y, rcond=None)[0]
fitted_bwt = X @ coef
residual = y - fitted_bwt
prediction_error = fitted_bwt - y
print(pd.DataFrame({
    "bwt": y,
    "fitted": fitted_bwt.round(1),
    "residual": residual.round(1),
    "minus_prediction_error": (-prediction_error).round(1),
}).head())
#>       bwt  fitted  residual  minus_prediction_error
#> 0  2523.0  3175.7    -652.7                  -652.7
#> 1  2551.0  3056.1    -505.1                  -505.1
#> 2  2557.0  2834.7    -277.7                  -277.7
#> 3  2594.0  2848.0    -254.0                  -254.0
#> 4  2600.0  2843.5    -243.5                  -243.5
print(np.max(np.abs(residual - (-prediction_error))))
#> 0.0
```

The last two columns agree in every row, and the largest difference between them over all 189 babies is 0, as Corollary 0.1 says.

Remark 0.1 (Prediction errors and residuals)

Remark 0.1 (Prediction errors and residuals). The prediction in Definition 0.10 can be a fitted value, for an observation used to fit the model, or a prediction for an observation that was not used to fit the model. Prediction error is oriented the same way as estimation error: estimate minus true value.

$$e = \hat{y} - y \quad \text{(prediction error)}$$
$$\varepsilon\left(\hat{\theta}\right) = \hat{\theta} - \theta \quad \text{(estimation error)}$$

For an observation used to fit the model, the prediction error is the negative of the residual (Theorem 0.1), so the two differ only in orientation.

Definition 0.11 (Mean squared error of predictions)

Definition 0.11 (Mean squared error of predictions)**.** The **mean squared error** of predictions $\widehat{y}_1, \ldots, \widehat{y}_n$ of observed outcomes $y_1, \ldots, y_n$ is the mean of their squared prediction errors:

$$\mathrm{MSE}(\widehat{y}) \stackrel{\text{def}}{=} \frac{1}{n} \sum_{i=1}^{n} e_i^2.$$

Remark 0.2 (Prediction MSE, estimator MSE and RSS)

Remark 0.2 (Prediction MSE, estimator MSE and RSS). Definition 0.11 averages squared errors over observations, where the mean squared error of an estimator takes an expectation over repeated samples. On the data used to fit the model, the mean squared error of the fitted values is the residual sum of squares divided by n (mean squared error and RSS):

$$\mathrm{MSE}(\widehat{y}) = \frac{1}{n}\mathrm{RSS}(\hat{\tilde{\theta}}), \qquad \mathrm{RSS}(\hat{\tilde{\theta}}) = n\ \mathrm{MSE}(\widehat{y}).$$

Example 0.9 (Fitted values, residuals, and prediction errors for three points)

Example 0.9 (Fitted values, residuals, and prediction errors for three points)**.** Take the points $(0, 1)$, $(1, 2)$, $(2, 2)$ and the line with $\beta_0 = 1$ and $\beta_x = 0.5$ from the residual sum of squares example, and treat the line as the fitted model.

x_i	y_i	$\widehat{y}_i = 1 + 0.5x_i$	$r_i = y_i - \widehat{y}_i$	$e_i = \widehat{y}_i - y_i$
0	1	1	0	0
1	2	1.5	0.5	−0.5
2	2	2	0	0

> Figure 1 shows the observed outcomes, the fitted values on the line, and the residual of the second point.

```r
fre_points <- tibble::tibble(x = c(0, 1, 2), y = c(1, 2, 2)) |>
  dplyr::mutate(fitted = 1 + 0.5 * x)

ggplot2::ggplot(fre_points, ggplot2::aes(x = x)) +
  ggplot2::geom_abline(intercept = 1, slope = 0.5) +
  ggplot2::geom_segment(
    ggplot2::aes(y = fitted, xend = x, yend = y),
    linetype = "dashed",
    color = "firebrick"
  ) +
  ggplot2::geom_point(ggplot2::aes(y = fitted), shape = 1, size = 3) +
  ggplot2::geom_point(ggplot2::aes(y = y), size = 3) +
  ggplot2::annotate(
    "text",
    x = 1.1, y = 1.75, hjust = 0,
    label = "r[2] == 0.5", parse = TRUE
  ) +
  ggplot2::labs(y = "y")
```

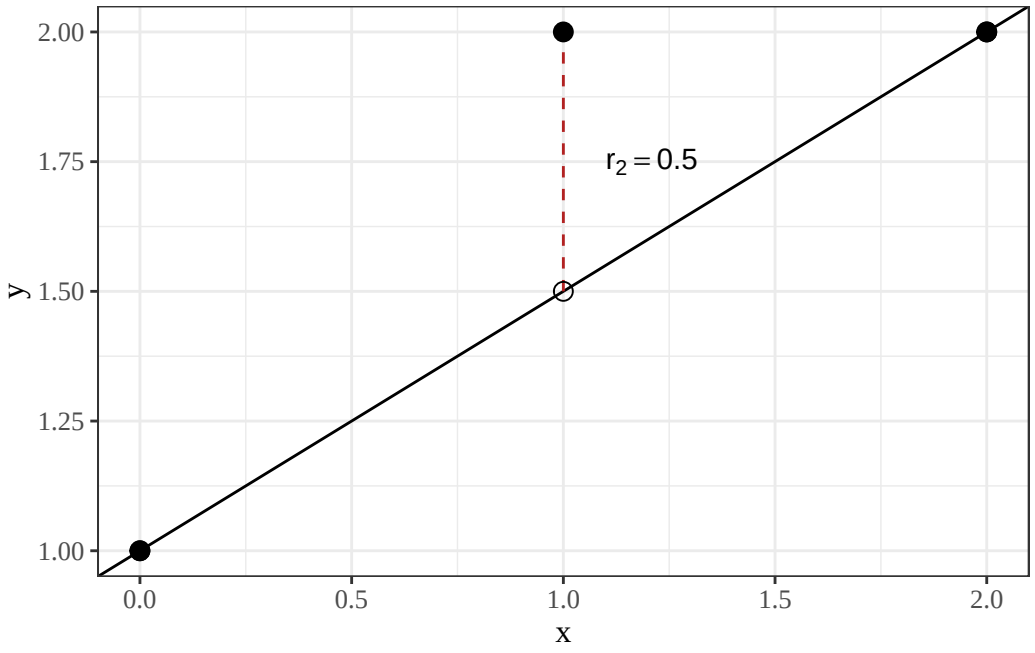


Figure 1: Observed outcomes (filled), fitted values on the line (open), and the residual of the second point (dashed), which points up from its fitted value to its observed outcome

The mean squared error of these predictions is

$$\begin{aligned}\mathrm{MSE}(\hat{y}) &= \frac{0^2 + (-0.5)^2 + 0^2}{3} \\ &= \frac{0.25}{3} \\ &\approx 0.083,\end{aligned}$$

the residual sum of squares, 0.25, divided by $n = 3$.

Bias

Definition 0.12 (Bias)

Definition 0.12 (Bias)**.** The **bias** of an estimator $\hat{\theta}$ for an estimand θ is the expected value of the estimation error:

$$\mathrm{Bias}\left(\hat{\theta}\right) \stackrel{\text{def}}{=} \mathrm{E}\left[\varepsilon\left(\hat{\theta}\right)\right] \tag{1}$$

Exercise 0.2 (Bias in terms of the expectation)

Exercise 0.2 (Bias in terms of the expectation)**.** Write the bias $\mathrm{Bias}\left(\hat{\theta}\right)$ of an estimator $\hat{\theta}$ in terms of its expected value $\mathrm{E}\left[\hat{\theta}\right]$ and the estimand θ.

Solution 0.2

Solution 0.2.

$$\begin{aligned}\mathrm{Bias}\left(\hat{\theta}\right) &\stackrel{\text{def}}{=} \mathrm{E}\left[\varepsilon\left(\hat{\theta}\right)\right] && \text{(definition of bias)} \\ &= \mathrm{E}\left[\hat{\theta} - \theta\right] && \text{(definition of estimation error)} \\ &= \mathrm{E}\left[\hat{\theta}\right] - \mathrm{E}[\theta] && \text{(linearity of expectation)} \\ &= \mathrm{E}\left[\hat{\theta}\right] - \theta && (\theta \text{ is a constant})\end{aligned}$$

Theorem 0.2 (Bias equals expectation minus truth)

Theorem 0.2 (Bias equals expectation minus truth)**.**

$$\mathrm{Bias}\left(\hat{\theta}\right) = \mathrm{E}\left[\hat{\theta}\right] - \theta$$

Proof

Proof. This is the solution to Exercise 0.2. □

Example 0.10 (Bias of two estimators of a mean)

Example 0.10 (Bias of two estimators of a mean)**.** Let $X_1, \dots, X_n$ each have expectation $\mathrm{E}[X_i] = \mu$, as in Example 0.4. By Theorem 0.2, the bias of the sample mean $\bar{X}$ is:

$$\begin{aligned}
\mathrm{Bias}\big(\bar{X}\big) &= \mathrm{E}\big[\bar{X}\big] - \mu && \text{(bias equals expectation minus truth)} \\
&= \mathrm{E}\left[\frac{1}{n}\sum_{i=1}^{n} X_i\right] - \mu && \text{(definition of } \bar{X}\text{)} \\
&= \frac{1}{n}\,\mathrm{E}\left[\sum_{i=1}^{n} X_i\right] - \mu && \text{(factor the constant } \tfrac{1}{n} \text{ out of the expectation)} \\
&= \frac{1}{n}\sum_{i=1}^{n} \mathrm{E}[X_i] - \mu && \text{(linearity of expectation)} \\
&= \frac{1}{n}\sum_{i=1}^{n} \mu - \mu && (\mathrm{E}[X_i] = \mu \text{ for every } i) \\
&= \frac{1}{n} \cdot n\mu - \mu && \text{(sum of } n \text{ copies of } \mu\text{)} \\
&= \mu - \mu && \text{(cancel } n\text{)} \\
&= 0 && \text{(subtract)}
\end{aligned}$$

and the bias of the single-observation estimator $\hat{\mu}^{(2)} = X_1$ is:

$$\begin{aligned}
\mathrm{Bias}(X_1) &= \mathrm{E}[X_1] - \mu && \text{(bias equals expectation minus truth)} \\
&= \mu - \mu && (\mathrm{E}[X_1] = \mu) \\
&= 0 && \text{(subtract)}
\end{aligned}$$

So both estimators have zero bias, even though $\bar{X}$ uses all n observations and X_1 uses only one.

Mean squared error

Definition 0.13 (Mean squared error)

Definition 0.13 (Mean squared error)**.** The **mean squared error** of an estimator $\hat{\theta}$, denoted $\mathrm{MSE}\big(\hat{\theta}\big)$, is the expectation of the square of the estimation error:

$$\mathrm{MSE}\big(\hat{\theta}\big) \stackrel{\text{def}}{=} \mathrm{E}\Big[\big(\varepsilon\big(\hat{\theta}\big)\big)^2\Big]$$

Exercise 0.3 (Expanding the squared bias)

Exercise 0.3 (Expanding the squared bias)**.** Using Theorem 0.2, expand the squared bias $\big(\mathrm{Bias}\big(\hat{\theta}\big)\big)^2$ in terms of $\mathrm{E}\big[\hat{\theta}\big]$ and the estimand θ.

Solution 0.3

Solution 0.3.

$$\begin{aligned}
\big(\mathrm{Bias}\big(\hat{\theta}\big)\big)^2 &= \big(\mathrm{E}\big[\hat{\theta}\big] - \theta\big)^2 && \text{(bias equals expectation minus truth)}\\
&= \big(\mathrm{E}\big[\hat{\theta}\big]\big)^2 - 2\,\mathrm{E}\big[\hat{\theta}\big]\theta + \theta^2 && \text{(expand the binomial square)}
\end{aligned}$$

Exercise 0.4 (Squared bias plus variance)

Exercise 0.4 (Squared bias plus variance)**.** Using Exercise 0.3, write $\big(\mathrm{Bias}\big(\hat{\theta}\big)\big)^2 + \mathrm{Var}\big(\hat{\theta}\big)$ in terms of $\mathrm{E}\big[\hat{\theta}^2\big]$, $\mathrm{E}\big[\hat{\theta}\big]$, and the estimand θ.

Solution 0.4

Solution 0.4. The variance is (simplified expression for variance):

$$\mathrm{Var}\big(\hat{\theta}\big) = \mathrm{E}\big[\hat{\theta}^2\big] - \big(\mathrm{E}\big[\hat{\theta}\big]\big)^2 \tag{2}$$

Add it to the squared bias from Exercise 0.3 and simplify:

$$\begin{aligned}
\big(\mathrm{Bias}\big(\hat{\theta}\big)\big)^2 + \mathrm{Var}\big(\hat{\theta}\big) &= \big(\mathrm{E}\big[\hat{\theta}\big]\big)^2 - 2\,\mathrm{E}\big[\hat{\theta}\big]\theta + \theta^2 + \mathrm{Var}\big(\hat{\theta}\big) && \text{(substitute the squared bias)}\\
&= \big(\mathrm{E}\big[\hat{\theta}\big]\big)^2 - 2\,\mathrm{E}\big[\hat{\theta}\big]\theta + \theta^2 + \mathrm{E}\big[\hat{\theta}^2\big] - \big(\mathrm{E}\big[\hat{\theta}\big]\big)^2 && \text{(substitute the variance)}\\
&= \mathrm{E}\big[\hat{\theta}^2\big] - 2\,\mathrm{E}\big[\hat{\theta}\big]\theta + \theta^2 + \big(\mathrm{E}\big[\hat{\theta}\big]\big)^2 - \big(\mathrm{E}\big[\hat{\theta}\big]\big)^2 && \text{(reorder the terms)}\\
&= \mathrm{E}\big[\hat{\theta}^2\big] - 2\,\mathrm{E}\big[\hat{\theta}\big]\theta + \theta^2 + \Big(\big(\mathrm{E}\big[\hat{\theta}\big]\big)^2 - \big(\mathrm{E}\big[\hat{\theta}\big]\big)^2\Big) && \text{(group the last two terms)}\\
&= \mathrm{E}\big[\hat{\theta}^2\big] - 2\,\mathrm{E}\big[\hat{\theta}\big]\theta + \theta^2 + 0 && (a - a = 0)\\
&= \mathrm{E}\big[\hat{\theta}^2\big] - 2\,\mathrm{E}\big[\hat{\theta}\big]\theta + \theta^2 && (a + 0 = a)
\end{aligned} \tag{3}$$

Exercise 0.5 (Expanding the mean squared error)

Exercise 0.5 (Expanding the mean squared error)**.** Write the mean squared error $\mathrm{MSE}\left(\hat{\theta}\right)$ in terms of $\mathrm{E}\left[\hat{\theta}^2\right]$, $\mathrm{E}\left[\hat{\theta}\right]$, and the estimand θ.

Solution 0.5

Solution 0.5.

$$\begin{aligned}
\mathrm{MSE}\left(\hat{\theta}\right) &\overset{\text{def}}{=} \mathrm{E}\left[\left(\varepsilon\left(\hat{\theta}\right)\right)^2\right] && \text{(definition of MSE)} \\
&= \mathrm{E}\left[(\hat{\theta}-\theta)^2\right] && \text{(definition of estimation error)} \\
&= \mathrm{E}\left[\hat{\theta}^2 - 2\hat{\theta}\theta + \theta^2\right] && \text{(expand the binomial square)} \\
&= \mathrm{E}\left[\hat{\theta}^2\right] - \mathrm{E}\left[2\hat{\theta}\theta\right] + \mathrm{E}[\theta^2] && \text{(linearity of expectation)} \\
&= \mathrm{E}\left[\hat{\theta}^2\right] - 2\,\mathrm{E}\left[\hat{\theta}\right]\theta + \mathrm{E}[\theta^2] && \text{(factor the constant } 2\theta \text{ out of the middle expectation)} \\
&= \mathrm{E}\left[\hat{\theta}^2\right] - 2\,\mathrm{E}\left[\hat{\theta}\right]\theta + \theta^2 && \text{(the expectation of a constant is that constant)}
\end{aligned}$$

Theorem 0.3 (Mean squared error equals bias squared plus variance)

Theorem 0.3 (Mean squared error equals bias squared plus variance)**.** *For any one-dimensional estimator* $\hat{\theta}$*:*

$$\mathrm{MSE}\left(\hat{\theta}\right) = \left(\mathrm{Bias}\left(\hat{\theta}\right)\right)^2 + \mathrm{Var}\left(\hat{\theta}\right) \tag{4}$$

Proof

Proof. By Exercise 0.5 and Exercise 0.4, $\mathrm{MSE}\left(\hat{\theta}\right)$ and $\left(\mathrm{Bias}\left(\hat{\theta}\right)\right)^2 + \mathrm{Var}\left(\hat{\theta}\right)$ both equal $\mathrm{E}\left[\hat{\theta}^2\right] - 2\,\mathrm{E}\left[\hat{\theta}\right]\theta + \theta^2$, so they are equal to each other. □

Example 0.11 (Mean squared error of two estimators of a mean)

Example 0.11 (Mean squared error of two estimators of a mean)**.** Continuing Example 0.10, suppose also that $X_1, \ldots, X_n$ are mutually independent, each with variance $\mathrm{Var}(X_i) = \sigma^2$. Both $\bar{X}$ and X_1 have zero bias, so by Theorem 0.3 each estimator's mean squared error equals its variance.
For X_1,

$$\begin{aligned}\mathrm{MSE}(X_1) &= 0^2 + \mathrm{Var}(X_1) \\ &= \sigma^2.\end{aligned}$$

For $\bar{X}$:

$$\begin{aligned}
\mathrm{MSE}(\bar{X}) &= \big(\mathrm{Bias}(\bar{X})\big)^2 + \mathrm{Var}(\bar{X}) && \text{(MSE equals bias squared plus variance)} \\
&= 0^2 + \mathrm{Var}(\bar{X}) && \text{(zero bias)} \\
&= \mathrm{Var}(\bar{X}) && (0^2 = 0) \\
&= \mathrm{Var}\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) && \text{(definition of } \bar{X}) \\
&= \frac{1}{n^2}\sum_{i=1}^{n} \mathrm{Var}(X_i) && \text{(variance of a linear combination of independent variables)} \\
&= \frac{1}{n^2}\sum_{i=1}^{n} \sigma^2 && (\mathrm{Var}(X_i) = \sigma^2 \text{ for every } i) \\
&= \frac{1}{n^2} \cdot n\sigma^2 && \text{(sum of } n \text{ copies of } \sigma^2) \\
&= \frac{\sigma^2}{n} && \text{(cancel one factor of } n)
\end{aligned}$$

The step to $\frac{1}{n^2}\sum_{i=1}^{n} \mathrm{Var}(X_i)$ uses the variance of a linear combination, whose covariance terms are all zero for independent variables.
So for any sample size $n > 1$,

$$\begin{aligned}\mathrm{MSE}(\bar{X}) &= \sigma^2/n \\ &< \sigma^2 \\ &= \mathrm{MSE}(X_1) :\end{aligned}$$

by mean squared error, the sample mean is the more accurate estimator. With $n = 50$ students, as in Example 0.4, the sample mean's mean squared error is 1/50 of X_1's.

Definition 0.14 (Root mean squared error)

Definition 0.14 (Root mean squared error)**.** The **root mean squared error** of an estimator $\hat{\theta}$ is the square root of its mean squared error:

$$\mathrm{RMSE}\big(\hat{\theta}\big) \stackrel{\text{def}}{=} \sqrt{\mathrm{MSE}\big(\hat{\theta}\big)}$$

It has the same units as the estimand. In Example 0.11, the root mean squared error of $\bar{X}$ is $\sigma/\sqrt{n}$.

Unbiased estimators

Definition 0.15 (Unbiased estimator)

Definition 0.15 (Unbiased estimator)**.** An estimator $\hat{\theta}$ is **unbiased** if its bias is zero:

$$\operatorname{Bias}\left(\hat{\theta}\right) = 0$$

An estimator whose bias is not zero is **biased**.

Example 0.12 (The sample mean is unbiased)

Example 0.12 (The sample mean is unbiased)**.** By Example 0.10, $\operatorname{Bias}\left(\bar{X}\right) = 0$ and $\operatorname{Bias}(X_1) = 0$, so both the sample mean $\bar{X}$ and the single-observation estimator X_1 are unbiased estimators of μ.

Example 0.13 (A biased estimator of the variance)

Example 0.13 (A biased estimator of the variance)**.** Let $X_1, \ldots, X_n$ be mutually independent, each with expectation μ and variance σ^2, and let $n \geq 2$. Consider two estimators of σ^2:

- the sample variance $S^2 \stackrel{\text{def}}{=} \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2$;
- the divide-by-n estimator $\hat{\sigma}^2 \stackrel{\text{def}}{=} \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2$.

Both estimators are built from the sum of squared deviations $\sum_{i=1}^{n}(X_i - \bar{X})^2$, so we first find its expectation. Using

$$\begin{aligned}\mathrm{E}[X_i^2] &= \operatorname{Var}(X_i) + (\mathrm{E}[X_i])^2 \\ &= \sigma^2 + \mu^2\end{aligned}$$

and

$$\begin{aligned}\mathrm{E}\left[\bar{X}^2\right] &= \operatorname{Var}\left(\bar{X}\right) + \left(\mathrm{E}\left[\bar{X}\right]\right)^2 \\ &= \sigma^2/n + \mu^2\end{aligned}$$

(by Example 0.11 and Example 0.10):

$$
\begin{aligned}
\mathrm{E}\left[\sum_{i=1}^{n}(X_i-\bar{X})^2\right] &= \mathrm{E}\left[\sum_{i=1}^{n}\left(X_i^2-2X_i\bar{X}+\bar{X}^2\right)\right] && \text{(expand each square)}\\
&= \mathrm{E}\left[\sum_{i=1}^{n}X_i^2-\sum_{i=1}^{n}2X_i\bar{X}+\sum_{i=1}^{n}\bar{X}^2\right] && \text{(split the sum)}\\
&= \mathrm{E}\left[\sum_{i=1}^{n}X_i^2-2\bar{X}\sum_{i=1}^{n}X_i+\sum_{i=1}^{n}\bar{X}^2\right] && \text{(factor } 2\bar{X} \text{ out of the middle sum)}\\
&= \mathrm{E}\left[\sum_{i=1}^{n}X_i^2-2\bar{X}\sum_{i=1}^{n}X_i+n\bar{X}^2\right] && \text{(sum of } n \text{ copies of } \bar{X}^2)\\
&= \mathrm{E}\left[\sum_{i=1}^{n}X_i^2-2\bar{X}\cdot n\bar{X}+n\bar{X}^2\right] && \left(\textstyle\sum_{i=1}^{n}X_i=n\bar{X}\right)\\
&= \mathrm{E}\left[\sum_{i=1}^{n}X_i^2-2n\bar{X}^2+n\bar{X}^2\right] && \text{(multiply } 2\bar{X}\cdot n\bar{X})\\
&= \mathrm{E}\left[\sum_{i=1}^{n}X_i^2-n\bar{X}^2\right] && \text{(collect the } \bar{X}^2 \text{ terms)}\\
&= \mathrm{E}\left[\sum_{i=1}^{n}X_i^2\right]-\mathrm{E}\left[n\bar{X}^2\right] && \text{(linearity of expectation)}\\
&= \sum_{i=1}^{n}\mathrm{E}[X_i^2]-\mathrm{E}\left[n\bar{X}^2\right] && \text{(linearity of expectation)}\\
&= \sum_{i=1}^{n}\mathrm{E}[X_i^2]-n\,\mathrm{E}\left[\bar{X}^2\right] && \text{(factor the constant } n \text{ out of the expectation)}\\
&= \sum_{i=1}^{n}(\sigma^2+\mu^2)-n\,\mathrm{E}\left[\bar{X}^2\right] && \text{(substitute } \mathrm{E}[X_i^2])\\
&= n(\sigma^2+\mu^2)-n\,\mathrm{E}\left[\bar{X}^2\right] && \text{(sum of } n \text{ copies of } \sigma^2+\mu^2)\\
&= n(\sigma^2+\mu^2)-n\left(\frac{\sigma^2}{n}+\mu^2\right) && \text{(substitute } \mathrm{E}\left[\bar{X}^2\right])\\
&= n\sigma^2+n\mu^2-n\left(\frac{\sigma^2}{n}+\mu^2\right) && \text{(distribute } n \text{ over } \sigma^2+\mu^2)\\
&= n\sigma^2+n\mu^2-\left(n\cdot\frac{\sigma^2}{n}+n\mu^2\right) && \text{(distribute } n \text{ over } \frac{\sigma^2}{n}+\mu^2)\\
&= n\sigma^2+n\mu^2-(\sigma^2+n\mu^2) && \text{(cancel } n \text{ in } n\cdot\frac{\sigma^2}{n})\\
&= n\sigma^2+n\mu^2-\sigma^2-n\mu^2 && \text{(distribute the minus sign)}\\
&= n\sigma^2-\sigma^2 && \text{(cancel } n\mu^2)\\
&= (n-1)\sigma^2 && \text{(factor out } \sigma^2)
\end{aligned}
$$

So

$$\begin{aligned}\mathrm{E}[S^2] &= \frac{1}{n-1}(n-1)\sigma^2 \\ &= \sigma^2,\end{aligned}$$

and by Theorem 0.2, $\mathrm{Bias}(S^2) = 0$: the sample variance is unbiased.
For the divide-by-n estimator, $\mathrm{E}[\hat{\sigma}^2] = \frac{1}{n}(n-1)\sigma^2$, so:

$$\begin{aligned}
\mathrm{Bias}(\hat{\sigma}^2) &= \mathrm{E}[\hat{\sigma}^2] - \sigma^2 && \text{(bias equals expectation minus truth)} \\
&= \frac{n-1}{n}\sigma^2 - \sigma^2 && \text{(substitute } \mathrm{E}[\hat{\sigma}^2]\text{)} \\
&= \frac{n-1}{n}\sigma^2 - \frac{n}{n}\sigma^2 && \text{(write } \sigma^2 \text{ over the common denominator } n\text{)} \\
&= \frac{n-1-n}{n}\sigma^2 && \text{(combine the fractions)} \\
&= \frac{-1}{n}\sigma^2 && (n-1-n=-1) \\
&= -\frac{\sigma^2}{n} && \text{(multiply)}
\end{aligned}$$

This estimator is biased: on average it underestimates σ^2, by an amount that shrinks to zero as n grows.

Exercise 0.6 (Expected value of an unbiased estimator)

Exercise 0.6 (Expected value of an unbiased estimator)**.** Using Theorem 0.2, find $\mathrm{E}\left[\hat{\theta}\right]$ for an unbiased estimator $\hat{\theta}$ of θ.

Solution 0.6

Solution 0.6.

$$\begin{aligned}
0 &= \mathrm{Bias}\left(\hat{\theta}\right) && \text{(definition of unbiased)} \\
&= \mathrm{E}\left[\hat{\theta}\right] - \theta && \text{(bias equals expectation minus truth)}
\end{aligned}$$

Adding θ to both sides gives $\mathrm{E}\left[\hat{\theta}\right] = \theta$.

Exercise 0.7 (Mean squared error of an unbiased estimator)

Exercise 0.7 (Mean squared error of an unbiased estimator)**.** Using Theorem 0.3, write the mean squared error of an unbiased estimator $\hat{\theta}$ in terms of its variance.

Solution 0.7

Solution 0.7.

$$\begin{aligned}
\mathrm{MSE}\left(\hat{\theta}\right) &= \left(\mathrm{Bias}\left(\hat{\theta}\right)\right)^2 + \mathrm{Var}\left(\hat{\theta}\right) && \text{(MSE equals bias squared plus variance)} \\
&= 0^2 + \mathrm{Var}\left(\hat{\theta}\right) && \text{(definition of unbiased)} \\
&= \mathrm{Var}\left(\hat{\theta}\right) && (0^2 = 0)
\end{aligned}$$

Theorem 0.4 (Properties of unbiased estimators)

Theorem 0.4 (Properties of unbiased estimators)**.** *If $\hat{\theta}$ is an unbiased estimator of θ, then:*

$$\mathrm{E}\left[\hat{\theta}\right] = \theta \tag{5}$$

$$\mathrm{MSE}\left(\hat{\theta}\right) = \mathrm{Var}\left(\hat{\theta}\right) \tag{6}$$

Proof

Proof. Equation 5 is the solution to Exercise 0.6, and Equation 6 is the solution to Exercise 0.7. □

Watch

5MinuteAI covers the properties of estimators, including bias, following the University of Washington's CSE 312 (5MinuteAI, n.d.).

Mean absolute error

Definition 0.16 (Mean absolute error)

Definition 0.16 (Mean absolute error)**.** The **mean absolute error** of an estimator is the expectation of the absolute value of the estimation error:

$$\mathrm{MAE}\left(\hat{\theta}\right) \stackrel{\text{def}}{=} \mathrm{E}\left[\left|\varepsilon\left(\hat{\theta}\right)\right|\right]$$

Example 0.14 (Mean absolute error of a Gaussian estimator)

Example 0.14 (Mean absolute error of a Gaussian estimator)**.** Suppose an estimator $\hat{\theta}$ has a Gaussian distribution with mean θ and standard deviation τ, so that its estimation error $\varepsilon\big(\hat{\theta}\big) = \hat{\theta} - \theta$ has the same distribution as τZ, where Z has a standard Gaussian distribution with density $\phi(z) = (2\pi)^{-1/2} e^{-z^2/2}$. Then:

$$
\begin{aligned}
\mathrm{MAE}\big(\hat{\theta}\big) &= \mathrm{E}[|\tau Z|] && \text{(definition of MAE)} \\
&= \mathrm{E}[\tau |Z|] && (|\tau Z| = \tau |Z| \text{ because } \tau > 0) \\
&= \tau\, \mathrm{E}[|Z|] && \text{(factor the constant } \tau \text{ out of the expectation)} \\
&= \tau \int_{-\infty}^{\infty} |z| \phi(z)\, dz && \text{(expectation of a function of } Z) \\
&= \tau \cdot 2 \int_{0}^{\infty} |z| \phi(z)\, dz && (|z|\phi(z) \text{ is symmetric about 0}) \\
&= \tau \cdot 2 \int_{0}^{\infty} z \phi(z)\, dz && (|z| = z \text{ for } z \geq 0) \\
&= \tau \cdot 2 [-\phi(z)]_{0}^{\infty} && (\phi'(z) = -z\phi(z)) \\
&= \tau \cdot 2 \Big(\lim_{z \to \infty} [-\phi(z)] - (-\phi(0)) \Big) && \text{(evaluate at the limits)} \\
&= \tau \cdot 2 \Big(\lim_{z \to \infty} [-\phi(z)] + \phi(0) \Big) && \text{(subtracting a negative is adding)} \\
&= \tau \cdot 2 (0 + \phi(0)) && (\phi(z) \to 0 \text{ as } z \to \infty) \\
&= \tau \cdot 2 \phi(0) && \text{(drop the zero term)} \\
&= \tau \cdot 2 (2\pi)^{-1/2} && (\phi(0) = (2\pi)^{-1/2}) \\
&= \tau \cdot 2 \cdot \frac{1}{\sqrt{2\pi}} && (\text{write } (2\pi)^{-1/2} \text{ as } 1/\sqrt{2\pi}) \\
&= \tau \cdot \frac{2}{\sqrt{2\pi}} && \text{(multiply)} \\
&= \tau \cdot \frac{\sqrt{4}}{\sqrt{2\pi}} && (\text{write } 2 \text{ as } \sqrt{4}) \\
&= \tau \sqrt{4/(2\pi)} && \text{(quotient of square roots)} \\
&= \tau \sqrt{2/\pi} && (\text{cancel the common factor 2 in } 4/(2\pi))
\end{aligned}
$$

So for an unbiased Gaussian estimator, the mean absolute error is about 0.80 times the standard deviation τ, while the root mean squared error is τ itself (Theorem 0.4).

Standard error

Definition 0.17 (Standard error)

Definition 0.17 (Standard error)**.** The **standard error** of an estimator $\hat{\theta}$ is the standard deviation of $\hat{\theta}$:

$$\mathrm{SE}\big(\hat{\theta}\big) \stackrel{\text{def}}{=} \mathrm{SD}\big(\hat{\theta}\big)$$

Example 0.15 (Standard error of the sample mean)

Example 0.15 (Standard error of the sample mean)**.** In Example 0.11, $\mathrm{Var}\big(\bar{X}\big) = \sigma^2/n$, so

$$\begin{aligned}\mathrm{SE}\big(\bar{X}\big) &= \sqrt{\sigma^2/n} \\ &= \sigma/\sqrt{n}.\end{aligned}$$

With $n = 50$ students, the standard error of the sample mean height is $\sigma/\sqrt{50} \approx 0.14\sigma$.

Exercise 0.8 (Spread of the estimation error)

Exercise 0.8 (Spread of the estimation error)**.** Show that the standard deviation of the estimation error $\varepsilon\big(\hat{\theta}\big)$ equals the standard error of $\hat{\theta}$.

Solution 0.8

Solution 0.8.

$$\begin{aligned}\mathrm{Var}\Big(\varepsilon\big(\hat{\theta}\big)\Big) &= \mathrm{Var}\big(\hat{\theta} - \theta\big) && \text{(definition of estimation error)} \\ &= \mathrm{Var}\big(\hat{\theta}\big) && \text{(subtracting a constant does not change a variance)}\end{aligned}$$

Taking square roots of both sides,

$$\begin{aligned}\mathrm{SD}\Big(\varepsilon\big(\hat{\theta}\big)\Big) &= \mathrm{SD}\big(\hat{\theta}\big) \\ &= \mathrm{SE}\big(\hat{\theta}\big).\end{aligned}$$

Theorem 0.5 (Standard error is the spread of the estimation error)

Theorem 0.5 (Standard error is the spread of the estimation error)**.**

$$\mathrm{SE}\big(\hat{\theta}\big) = \mathrm{SD}\big(\varepsilon\big(\hat{\theta}\big)\big)$$

Proof

Proof. This is the solution to Exercise 0.8. □

Remark 0.3 (The name "standard error")

Remark 0.3 (The name "standard error")*.* "Standard error" is a confusing name in two ways. It is defined through the estimator's own spread, not through the estimation error (although Theorem 0.5 shows that the two spreads are equal). It is also a synonym for the standard deviation of an estimator, so it can look redundant. The name persists because standard errors are the building blocks of p-values and confidence intervals, so they come up often enough to deserve their own name.

Exercise 0.9 (Squared standard error from MSE and bias)

Exercise 0.9 (Squared standard error from MSE and bias)**.** Using Theorem 0.3, write the squared standard error $\big(\mathrm{SE}\big(\hat{\theta}\big)\big)^2$ in terms of the mean squared error and the bias of $\hat{\theta}$.

Solution 0.9

Solution 0.9*.*

$$\begin{aligned}
\mathrm{MSE}\big(\hat{\theta}\big) &= \big(\mathrm{Bias}\big(\hat{\theta}\big)\big)^2 + \mathrm{Var}\big(\hat{\theta}\big) && \text{(MSE equals bias squared plus variance)}\\
\mathrm{Var}\big(\hat{\theta}\big) &= \mathrm{MSE}\big(\hat{\theta}\big) - \big(\mathrm{Bias}\big(\hat{\theta}\big)\big)^2 && \text{(subtract } \big(\mathrm{Bias}\big(\hat{\theta}\big)\big)^2 \text{ from both sides)}\\
\big(\mathrm{SD}\big(\hat{\theta}\big)\big)^2 &= \mathrm{MSE}\big(\hat{\theta}\big) - \big(\mathrm{Bias}\big(\hat{\theta}\big)\big)^2 && \big(\mathrm{Var}\big(\hat{\theta}\big) = \big(\mathrm{SD}\big(\hat{\theta}\big)\big)^2\big)\\
\big(\mathrm{SE}\big(\hat{\theta}\big)\big)^2 &= \mathrm{MSE}\big(\hat{\theta}\big) - \big(\mathrm{Bias}\big(\hat{\theta}\big)\big)^2 && \text{(definition of standard error: } \mathrm{SE}\big(\hat{\theta}\big) = \mathrm{SD}\big(\hat{\theta}\big)\big)
\end{aligned}$$

Corollary 0.2 (Standard error squared equals MSE minus squared bias)

Corollary 0.2 (Standard error squared equals MSE minus squared bias)**.** *The squared standard error is what remains of the mean squared error after the squared bias is removed:*

$$\big(\mathrm{SE}\big(\hat{\theta}\big)\big)^2 = \mathrm{MSE}\big(\hat{\theta}\big) - \big(\mathrm{Bias}\big(\hat{\theta}\big)\big)^2$$

Proof

Proof. This is the solution to Exercise 0.9. □

Exercise 0.10 (Standard error of an unbiased estimator)

Exercise 0.10 (Standard error of an unbiased estimator)**.** Using Equation 6, write the standard error of an unbiased estimator $\hat{\theta}$ in terms of its mean squared error.

Solution 0.10

Solution 0.10. By Equation 6, $\text{MSE}\big(\hat{\theta}\big) = \text{Var}\big(\hat{\theta}\big)$. Taking square roots of both sides,

$$\begin{aligned}\sqrt{\text{MSE}\big(\hat{\theta}\big)} &= \text{SD}\big(\hat{\theta}\big) \\ &= \text{SE}\big(\hat{\theta}\big).\end{aligned}$$

Corollary 0.3 (For unbiased estimators, SE equals root MSE)

Corollary 0.3 (For unbiased estimators, SE equals root MSE)**.** *If $\hat{\theta}$ is unbiased, then its standard error equals its root mean squared error:*

$$\text{SE}\big(\hat{\theta}\big) = \sqrt{\text{MSE}\big(\hat{\theta}\big)}$$

Proof

Proof. This is the solution to Exercise 0.10. □

References

5MinuteAI. n.d. *Probability & Statistics*. Video playlist on YouTube. Accessed October 12, 2026. https://www.youtube.com/playlist?list=PLeB45KifGiuHesi4PALNZSYZFhViVGQJK.

Box, George E. P., and Norman Richard. Draper. 1987. *Empirical Model-Building and Response Surfaces*. Wiley Series in Probability and Mathematical Statistics. Applied Probability and Statistics. Wiley.

Dunn, Peter K, and Gordon K Smyth. 2018. *Generalized Linear Models with Examples in R*. Vol. 53. Springer. https://doi.org/10.1007/978-1-4419-0118-7.

Hosmer, David W., and Stanley Lemeshow. 1989. *Applied Logistic Regression*. Wiley.

Lawrance, Rachael, Evgeny Degtyarev, Philip Griffiths, et al. 2020. "What Is an Estimand, and How Does It Relate to Quantifying the Effect of Treatment on Patient-Reported Quality of Life Outcomes in Clinical Trials?" *Journal of Patient-Reported Outcomes* 4 (1): 1–8. https://doi.org/10.1186/s41687-020-00218-5.

Pohl, Moritz, Lukas Baumann, Rouven Behnisch, Marietta Kirchner, Johannes Krisam, and Anja Sander. 2021. "Estimands—A Basic Element for Clinical Trials." *Deutsches Ärzteblatt International* 118 (51-52): 883–88. https://doi.org/10.3238/arztebl.m2021.0373.

Van Buuren, Stef. 2018. *Flexible Imputation of Missing Data*. CRC press. https://stefvanbuuren.name/fimd/.